J D COLLEGE OF ENGINEERING
& MANAGEMENT, NAGPUR

Secure File Sharing System Using Visual Cryptography and Blockchain

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Date of Presentation : 05/05/26

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INTRODUCTIONc

Problem Statement

- Traditional file sharing systems are vulnerable to data breaches and unauthorized access.
- Single-point encryption can be compromised, risking sensitive data.

Objective

- To develop a secure file sharing system using **Visual Cryptography**.
- To ensure data privacy by splitting files into multiple shares.
- To use **Blockchain** for integrity and access control.
- To provide a reliable and decentralized solution.

Project Idea

- The system splits a file into encrypted shares using VC.
- Shares are stored securely (IPFS).
- Blockchain stores hashes and manages access permissions.
- The original file is reconstructed only when valid shares are combined.

LITERATURE REVIEW

Work	What They Did	What's Good	What's Missing (Gap)
Moni Naor & Adi Shamir (1994)	Introduced Visual Cryptography to split images into shares	No decryption needed, very secure	Only works well for images, not full files
Adi Shamir (1979)	Proposed Secret Sharing (SSS) for splitting data mathematically	Strong security model	Needs computation to reconstruct
IPFS	Decentralized file storage using distributed nodes	No single point of failure	Data availability depends on nodes
Ethereum	Smart contracts for secure access control	Tamper-proof and transparent	Cost and scalability issues
Recent Systems	Combine VC + Blockchain + Storage	Better security + decentralization	Complex integration, not optimized

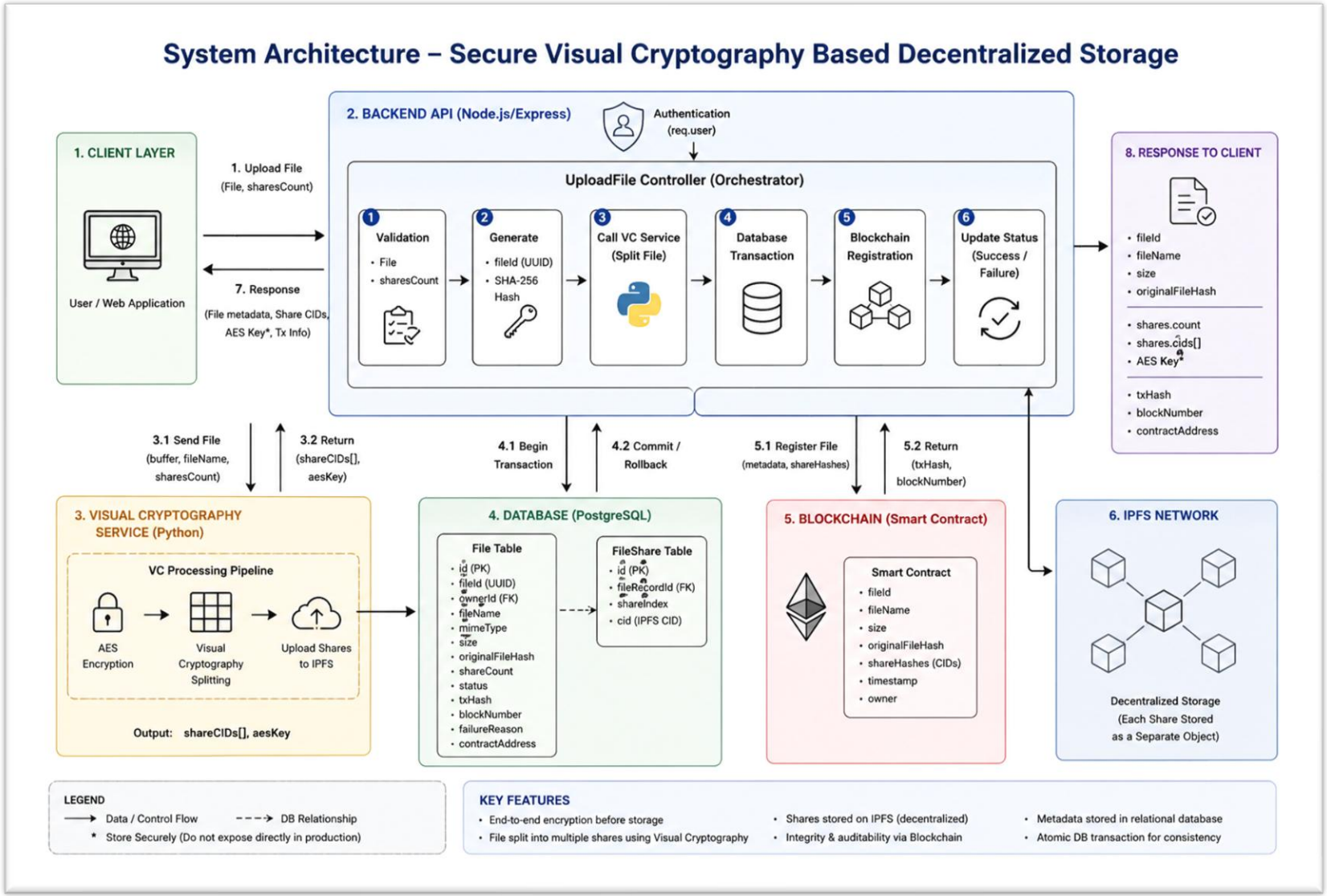
PROPOSED SYSTEM

- The system provides **secure file sharing** using **Visual Cryptography and Shamir's Secret Sharing**.
- A file is divided into multiple encrypted shares using a **Python-based VC service**.
- These shares are stored securely and cannot reveal any information individually.
- A **Node.js backend** manages file operations, share handling, and API communication.
- **PostgreSQL** is used as the database to store metadata, user data, and file references.
- Authentication is handled using **HTTP-only cookies**, ensuring secure session management.
- **Blockchain (smart contracts)** is used to store hashes and enforce access control.
- The original file is reconstructed only when authorized shares are combined.

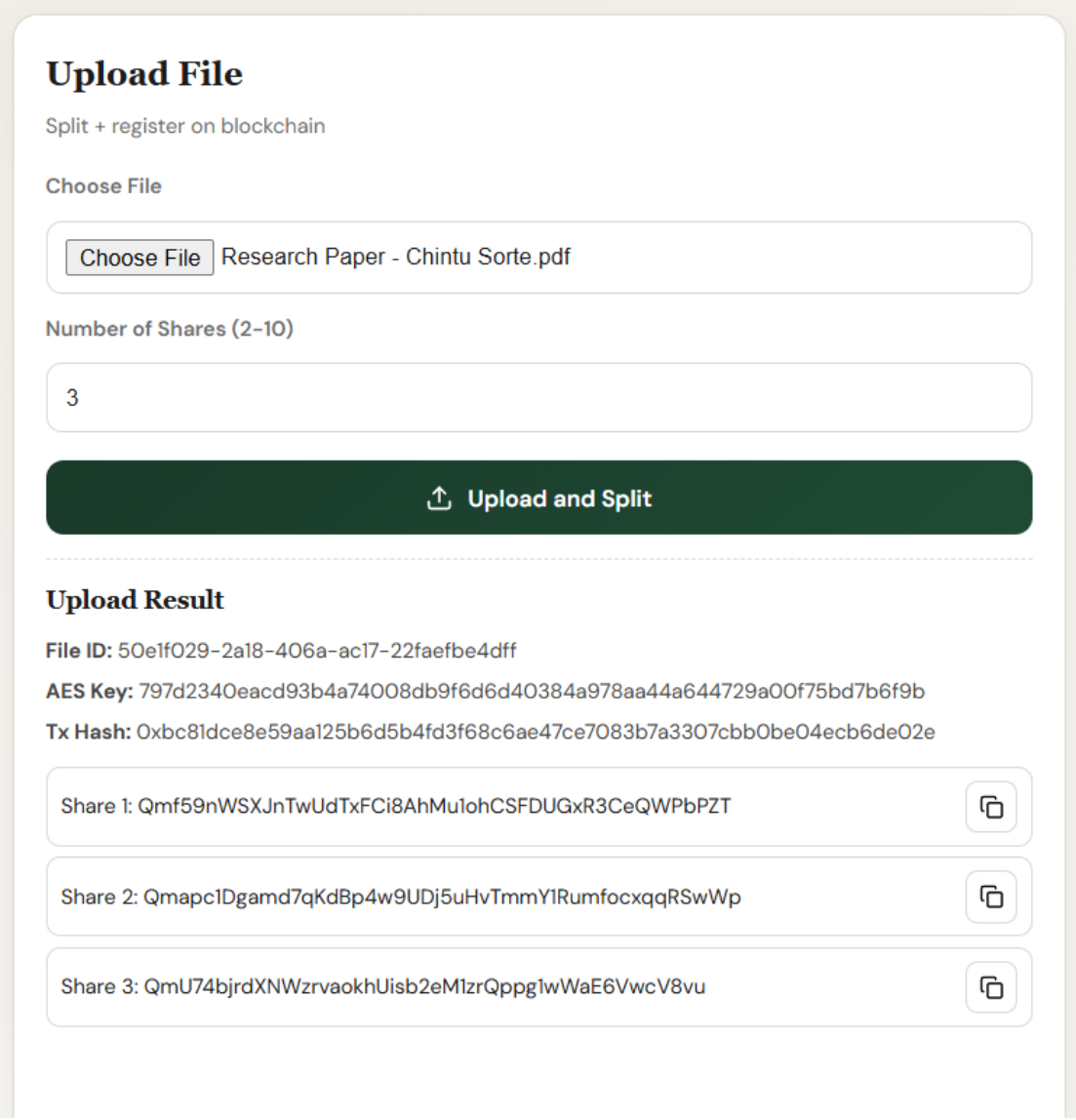
TECHNOLOGIES USED

- **Frontend:** React + TypeScript (Vite + Tailwind CSS)
- **Backend:** Node.js (Express)
- **VC Service:** Python (for share splitting & reconstruction)
- **Database:** PostgreSQL
- **Authentication:** HTTP-only Cookies (session-based auth)
- **Blockchain:** Smart Contracts (Polygon / Ethereum)
- **Storage (optional based on design):** IPFS / Local storage

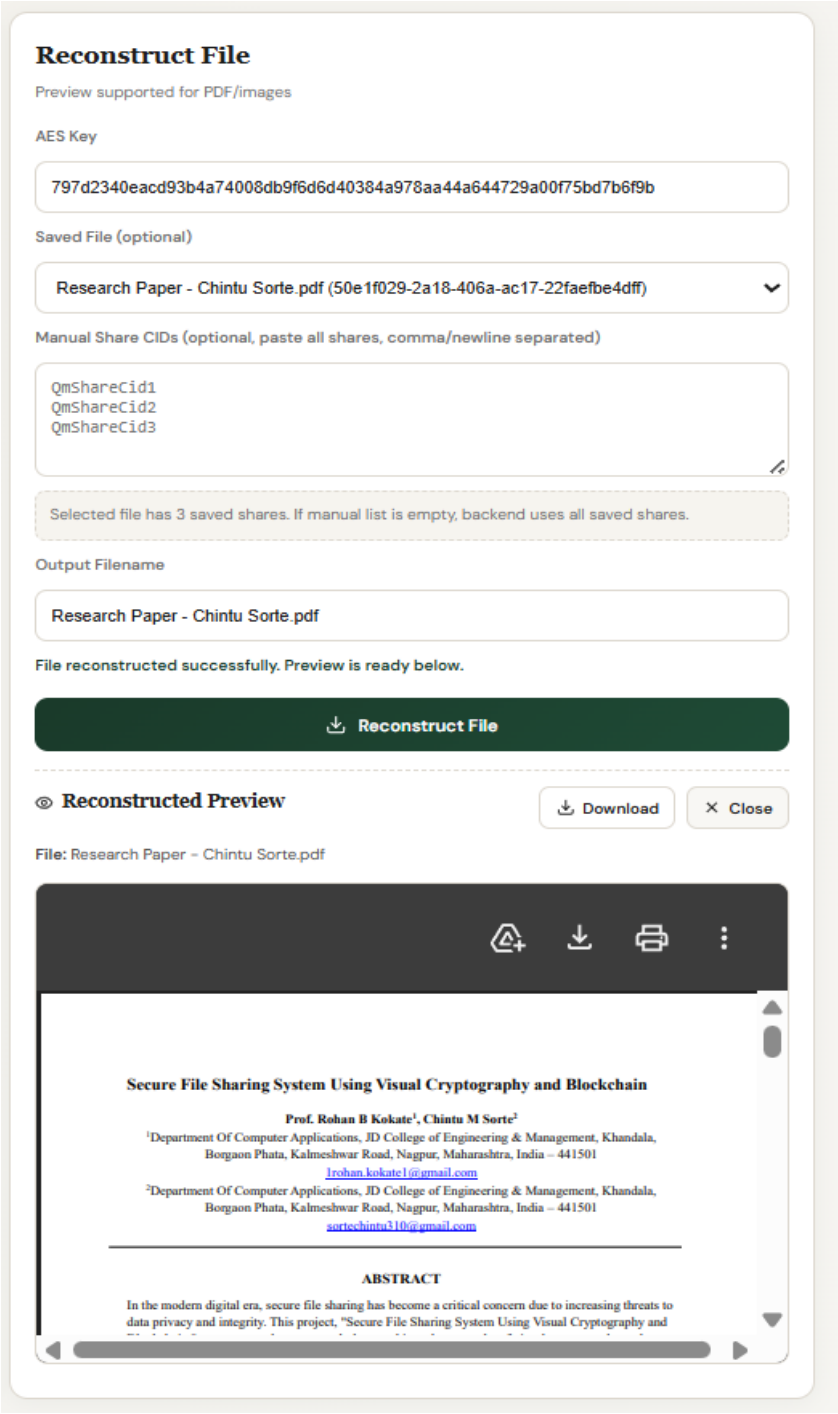
SYSTEM ARCHITECTURE



RESULTS & ANALYSIS



1. User uploads a file through the frontend interface.
2. The backend sends the file to the **Python VC service**.
3. The file is split into multiple encrypted shares.
4. Shares are stored securely and metadata is saved in PostgreSQL.
5. Blockchain stores hash for integrity verification.
6. **Result:** File is securely stored in distributed form (not readable individually).



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1. Authorized user requests the file.
2. Backend fetches required shares from storage.
3. Shares are sent to the Python VC service.
4. The original file is reconstructed using valid shares.
5. Integrity is verified using blockchain hash.
6. **Result:** File is restored accurately only for authorized users.

PROJECT TIMELINE

Phase	Work Done	Status
Problem Identification	Defined secure file sharing problem and objectives	Completed
Literature Review	Studied Visual Cryptography, Secret Sharing, Blockchain	Completed
System Design	Designed full architecture (Frontend, Backend, VC Service, Blockchain)	Completed
Backend Development	Developed APIs, authentication (HTTP-only cookies)	Completed
VC Service (Python)	Implemented share splitting and reconstruction logic	Completed
Database Setup	Designed schema using PostgreSQL	Completed
Frontend Development	Built user interface using React and TypeScript	Completed
Integration	Connected all components (Frontend, Backend, VC, Blockchain)	Completed
Testing	Performed functional and security testing	Completed
Deployment	Deployed system and finalized documentation	Completed

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THANKYOU